

Integration Worksheet: Pick the Method II

WS7.5.1

In groups of 3 or 4 people, examine each integral below and determine the best method to use on it. Give some detail about how you'd start.

1. $\int x \ln x \, dx$

9. $\int x^2 \cos x \, dx$

2. $\int \frac{x^3 + 5x}{3x^2} \, dx = \int \frac{x^3}{3x^2} + \frac{5x}{3x^2} \, dx$
 $= \int \frac{x}{3} + \frac{5}{3x} \, dx$

10. $\int \frac{\tan^2 x \csc x}{\sec x} \, dx$

3. $\int \frac{3x^2}{x^3 + 5x} \, dx$ P.F. $\frac{3x^2}{x(x^2+5)} = \frac{A}{x} + \frac{Bx+C}{x^2+5}$

11. $\int \arctan(x) \, dx$ IBP $= x \arctan x - \int \frac{x}{x^2+1} \, dx$
 $u = \arctan x \quad dv = dx$
 $du = \frac{1}{x^2+1} \, dx \quad v = x$
 $u = x^2+1$
 $du = 2x \, dx$

4. $\int \sin(x) \cos(\cos x) \, dx$

12. $\int \cos \sqrt{x} \, dx = \int \cos u \cdot \underbrace{2\sqrt{x}}_u \, du = 2 \int u \cos u \, du$
 $u = \sqrt{x} \, dx$
 $du = \frac{1}{2\sqrt{x}} \, dx$
 $2\sqrt{x} \, du = dx$
 Tabular IBP

5. $\int \frac{x}{\sqrt{x^2-4}} \, dx$ Trig sub
 $u = x^2-4$
 $du = 2x \, dx$
 $\int \frac{x}{\sqrt{u}} \cdot \frac{du}{2x} = \int \frac{1}{\sqrt{u}} \, du = \int u^{-1/2} \, du$

13. $\int \frac{x^3 + 3x + 5}{2x^3 - 5x^2 - 3x} \, dx$ PF
 $u = 2x^3 - 5x^2 - 3x$
 $du = 6x^2 - 10x - 3 \, dx$

6. $\int \frac{x^2}{\sqrt{x^2-4}} \, dx$ Trig sub
 $u = x^2-4 \quad du = 2x \, dx$ ← Does not work
 $\int \frac{x^2}{\sqrt{u}} \cdot \frac{du}{2x} = \int \frac{x}{2\sqrt{u}} \, du$

14. $\int \frac{6x^2 + 3x + 1}{4x^3 + 3x^2 + 2x} \, dx$ PF
 $u = 4x^3 + 3x^2 + 2x$
 $du = (12x^2 + 6x + 2) \, dx$

7. $\int \frac{x+4}{x^2+1} \, dx = \int \frac{x}{x^2+1} + \frac{4}{x^2+1} \, dx$
 $u = x^2+1 \quad u = \arctan(x)$

15. $\int \underbrace{3e^x}_{\text{constant}} + 5xe^x \, dx = 3e^x + \int 5xe^x \, dx$
 Tabular IBP

8. $\int \frac{x+4}{x^2-1} \, dx \quad \frac{A}{x-1} + \frac{B}{x+1}$
 $(x-1)(x+1)$

16. $\int \frac{\sin^3 x}{\cos x} \, dx$

3.0.3 Example:

$$\int \frac{2x^2 - x + 4}{x(x^2 + 4)} dx \quad \left[\frac{2x^2 - x + 4}{x(x^2 + 4)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 4} \right] \times (x^2 + 4)$$

$$2x^2 - x + 4 = A(x^2 + 4) + (Bx + C)x$$

$$= Ax^2 + 4A + Bx^2 + Cx$$

$$\underline{x=0}$$

$$4 = A(4)$$

$$A = 1$$

$$\frac{x^2}{2} = A + B$$

$$1 = B$$

$$\frac{x}{-1} = C$$

$$\int \frac{1}{x} + \int \frac{x}{x^2 + 4} - \int \frac{1}{x^2 + 4} dx$$

$$\left\{ \begin{array}{l} u = x^2 + 4 \\ du = 2x dx \\ \frac{1}{2} \int \frac{du}{u} \end{array} \right\}$$

$$\ln|x| + \frac{1}{2} \ln(x^2 + 4) - \frac{1}{2} \arctan\left(\frac{x}{2}\right) + C$$

1. IBP: $u = \ln x$ $dv = x dx$
2. Split into two fractions and integrate separately
3. partial fractions.
4. u -sub $u = \cos x$
5. u -sub $u = x^2 - 4$ is best. Trig sub $x = 2 \sec \theta$ also works but is a lot more work.
6. Trig sub $x = 2 \sec \theta$
7. Split into $\int \frac{x}{x^2+1} dx + \int \frac{4}{x^2+1} dx$, then $u = x^2 + 1$ and arctan, respectively
8. partial fractions
9. tabular IBP
10. Rewrite as sines and cosines and simplify.
11. IBP: $u = \tan^{-1} x$, $dv = dx$
12. u -sub $u = \sqrt{x}$, then IBP
13. partial fractions
14. u -sub $u = 4x^3 + 3x^2 + 2x$ is best. partial fractions also works.
15. Split. then $3e^e$ is a constant and $\int 5xe^x dx$ can be done with tabular IBP
16. u -sub $u = \cos x$